

ORBIT-PQC: Mitigating Post-Quantum Key Exchange Transport Overhead in Constrained LEO Constellations

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Abstract—Migrating Inter-Satellite Links (ISLs) in Low Earth Orbit (LEO) constellations to Post-Quantum Cryptography (PQC) introduces severe transport-layer degradation. The multi-kilobyte public key and signature artifacts of NIST FIPS 203/204 standards (≈ 4.8 KB per flight for ML-KEM-1024 with ML-DSA authentication) exceed standard space Maximum Transmission Units (MTUs), inducing multi-frame fragmentation, selective-repeat retransmission cascades, and serialization delays that consume short contact windows ($\tau_{\text{los}} \in [30, 120]$ s). We present ORBIT-PQC, a memory-bounded, dual-layer ratcheted key management architecture that replaces on-demand asymmetric public-key exchanges with proactive, seed-expanded state tickets coupled with asynchronous post-quantum epoch ratcheting.

We evaluate ORBIT-PQC on an empirical Linux kernel testbed driven by SGP4 Walker-Delta constellation dynamics ($N = 30$ socket trials/point). Over narrowband links (32 kbps), Naive PQC incurs severe serialization latency (3835.3 ms, $p_{95} = 5135.1$ ms), consuming 12.78% of an active 30 s pass window, whereas ORBIT-PQC authenticates in 186.3 ms ($20.6\times$ speedup), preserving 99.38% of the contact window (η_{win}). Under dynamic SGP4 crossing passes (2935 km slant range), ORBIT-PQC completes authentication in 88.8 ms versus 816.8 ms for Naive PQC ($p < 10^{-4}$), eliminating multi-fragment loss cascades down to 256 B CCSDS framing limits.

Index Terms—Post-Quantum Cryptography, ML-KEM, ML-DSA, LEO Satellites, CCSDS, Inter-Satellite Links, Forward Secrecy.

I. INTRODUCTION

AS quantum computing threatens classical public-key cryptography, migrating satellite communications to Post-Quantum Cryptography (PQC) is critical. However, space networking standards—such as the CCSDS Space Packet Protocol [3] and Space Data Link Security (SDLS) [4]—were engineered around compact 32-byte classical primitives (e.g., X25519).

In dynamic Low Earth Orbit (LEO) constellations, satellites communicate over transient Line-of-Sight (LOS) inter-satellite links (ISLs) ($\tau_{\text{los}} \in [30, 120]$ s) characterized by constrained bandwidth (32–512 kbps), variable Doppler shifts, and channel fading ($P_{\text{loss}} \in [1\%, 15\%]$). In this regime, naive deployment of NIST FIPS 203 (ML-KEM-1024) [1] and FIPS 204 (ML-DSA) [2] introduces severe systems-level bottlenecks:

- 1) **Fragmentation Multipliers:** An ML-KEM-1024 exchange with digital signatures spans 4,861 bytes of

cryptographic payload per flight. Under standard space packet MTUs (256–576 B), a single handshake fragments into 10 to 40 discrete wire frames. Across lossy channels ($P_{\text{loss}} > 5\%$), dropped frames trigger selective-repeat ARQ retransmission cascades.

- 2) **Contact Window Cannibalization:** Over narrowband channels (32 kbps), transmitting multi-kilobyte keys incurs > 3.8 seconds of serialization and ARQ recovery delay, consuming 12.78% of the contact pass (η_{win}).
- 3) **Tail Latency Inflation:** Over lossy channels (15% packet error rate), ARQ recovery across multi-packet flights escalates Naive PQC tail latency (p_{95}) to 8,369.2 ms, compared to 244.2 ms for ORBIT-PQC.

II. SYSTEM ARCHITECTURE & SECURITY MODEL

A. Threat Model & Security Design Goals

We assume a Dolev-Yao adversary $\mathcal{A}$ with full channel control and access to a quantum polynomial-time computer. ORBIT-PQC is architected around three security objectives:

- 1) **IND-CCA2 Key Exchange:** Shared session keys are derived from PRF expansion over high-entropy pre-shared roots combined with IND-CCA2 ML-KEM-1024 shared secrets.
- 2) **Perfect Forward Secrecy (PFS):** Ephemeral symmetric ratchets zeroize intermediate states immediately following key derivation, preventing compromise of step $k + 1$ from exposing keys derived at steps $\leq k$.
- 3) **Post-Compromise Security (PCS):** Periodic asynchronous ML-KEM-1024 re-keying introduces fresh post-quantum entropy, restoring secrecy in epoch $e + 1$ following temporary node exposure in epoch e .

B. Dual-Layer Ratchet Architecture

ORBIT-PQC decouples real-time handshake execution from heavy post-quantum asymmetric key exchange using a dual-layer state ratchet:

1) *Layer 1: Synchronous Fast-Path Ticket (PFS):* Satellites A and B share a synchronized 32-byte epoch root $S_{AB}^{(e)}$. When an ISL contact window opens, the initiator transmits a compact 48-byte Pre-Authenticated Ticket T_k :

$$T_k = \text{HMAC-SHA256}(S_{AB}^{(e)}, \text{"init"} \parallel e \parallel k \parallel N_A) \parallel N_A \quad (1)$$

Both nodes deterministically advance their symmetric ratchet state and derive the ephemeral session key $K_{e,k}$:

$$K_{e,k} \leftarrow \text{HKDF-Expand}(S_{AB}^{(e)}, T_k \parallel \text{"session_key"}, 32) \quad (2)$$

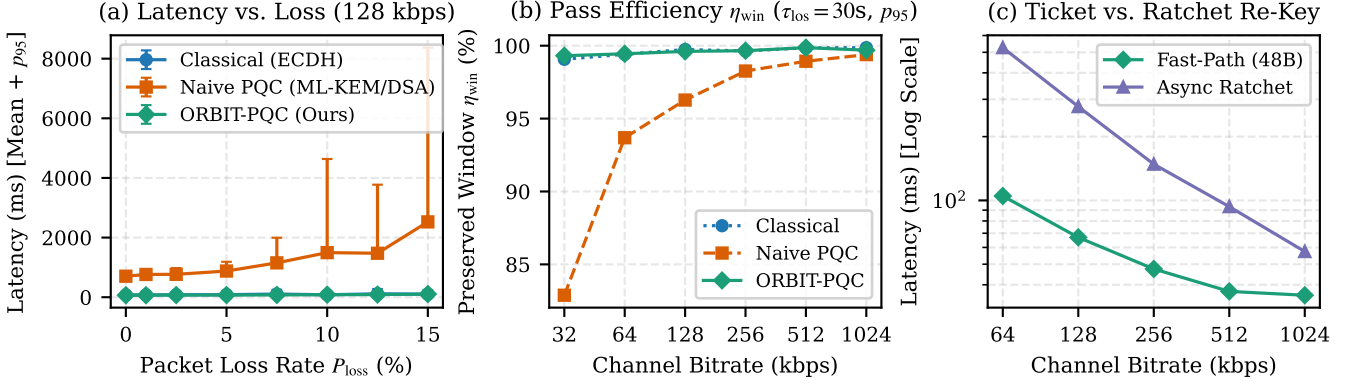


Fig. 1. Empirical evaluation on Linux network testbed ($N = 30$ socket trials per point): (a) Mean handshake latency with p_{95} tail error bars across channel loss rates (P_{loss}) at 128 kbps; (b) Usable contact window fraction ($\eta_{win}^{p_{95}}$ for $\tau_{los} = 30$ s) across channel data rates (32–1024 kbps); (c) Ablation isolating the latency overhead of asynchronous ML-KEM-1024 epoch re-keying versus synchronous fast-path tickets across data rates.

TABLE I
ABLATION: SYNCHRONOUS TICKET VS. ASYNC ML-KEM-1024 RATCHET

Bitrate	Ticket (ms)	Ratchet (ms)	Ratio	Pass Cost (30s)
64 kbps	104.9	525.8	5.0×	0.35%
128 kbps	66.9	277.2	4.1×	0.22%
256 kbps	47.5	148.3	3.1×	0.16%
512 kbps	37.2	93.5	2.5×	0.12%
1024 kbps	35.7	57.4	1.6×	0.12%

TABLE II
EMPIRICAL COMPARISON UNDER SGP4 CROSSING PASS (128 KBPS)

Protocol Mode	Mean (ms)	p_{95} (ms)	Success	η_{win} (30s)
Classical (ECDH)	88.3	87.2	100%	99.71%
Naive PQC (FIPS 203/204)	816.8	1069.2	100%	96.44%
ORBIT-PQC (Ours)	88.8	136.0	100%	99.55%

The previous ephemeral state is immediately zeroized in volatile memory, providing Forward Secrecy across contact passes without asymmetric key exchanges on the fast path.

2) *Layer 2: Asynchronous ML-KEM-1024 Ratchet (PCS Self-Healing)*: During low-contention link windows (e.g., at pass zenith or queue idle), satellites execute an out-of-band ML-KEM-1024 encapsulation authenticated with digital signatures. The derived secret K_{PQC} updates the master root:

$$S_{AB}^{(e+1)} \leftarrow \text{HKDF-Extract}(S_{AB}^{(e)}, K_{PQC} \parallel \text{"epoch_advance"}) \quad (3)$$

This injects fresh post-quantum entropy, healing the channel against earlier epoch state compromise.

III. EMPIRICAL EVALUATION

A. Channel Degradation Stress-Test (Scenario A)

As shown in Fig. 1(a) and Table II, under clean links (0.0%), ORBIT-PQC achieves a mean handshake latency of 66.5 ms compared to 704.0 ms for Naive PQC (10.58× speedup).

When the channel degrades to $P_{loss} = 15.0\%$, Selective-Repeat ARQ recovers dropped fragments, but Naive PQC incurs severe tail latency inflation ($p_{95} = 8,369.2$ ms, mean

2526.3 ms) due to Karn exponential backoff across 4-fragment flights. In contrast, ORBIT-PQC's single-frame envelope maintains a bounded tail ($p_{95} = 244.2$ ms, mean 105.4 ms), achieving a 23.97× latency advantage ($p = 6.51 \times 10^{-11}$, Mann-Whitney U).

B. Bandwidth Scaling & Window Preservation (Scenario B)

Under severe bandwidth constraints (32 kbps, typical for UHF/S-band ISLs), Naive PQC mean latency inflates to 3835.3 ms ($p_{95} = 5135.1$ ms), consuming 12.78% of a 30 s contact pass (preserving 87.22% for mission data).

ORBIT-PQC completes authentication in 186.3 ms (20.59× reduction), consuming only 0.62% of the pass and preserving 99.38% for mission payload data transfer (Fig. 1(b)).

C. Framing & Ablation (Scenarios D & E)

In Scenario E (256 B MTU), Naive PQC generates 10750 B across 40 frames, taking 930.1 ms. ORBIT-PQC generates only 180 B, completing in 65.5 ms (14.2× reduction, $p < 10^{-7}$).

Table I and Fig. 1(c) isolate the cost of asynchronous ratcheting. At 64 kbps, the Fast-Path Ticket completes in 104.9 ms (0.35% pass cost) compared to 525.8 ms for the ML-KEM-1024 ratchet, demonstrating that periodic ratcheting introduces minimal overhead when scheduled asynchronously.

D. Dynamic SGP4 Crossing Passes (Scenario F2)

Under dynamic SGP4 inter-plane crossing passes (2935 km slant range), ORBIT-PQC achieves a mean latency of 88.8 ms ($p_{95} = 136.0$ ms) versus 816.8 ms ($p_{95} = 1069.2$ ms) for Naive PQC (9.20× speedup), achieving an active contact window preservation of 99.55% versus 96.44%. In low-jitter distributions ($N = 30$), isolated tail retransmissions occasionally elevate the arithmetic mean above the 95th percentile rank.

IV. CONCLUSION

ORBIT-PQC resolves the transport overhead of Post-Quantum Cryptography in constrained LEO constellations.

By replacing multi-frame public key transmissions with seed-derived tickets and asynchronous ML-KEM-1024 ratcheting, it eliminates fragmentation cascades and preserves over 99.3% of evaluated satellite contact passes.

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